INDUSTRY FIELD NOTE

# China's AI stack localizes under constraint

Emerging signal · Evidence current to 27 August 2026

## Executive thesis

China's AI stack is localizing under constraint. Compute, memory, advanced packaging, model efficiency, cloud delivery, power and policy are increasingly interdependent because weakness in one layer limits the usefulness of progress in another. The near-term signal is clear: companies and policymakers are investing across the stack rather than relying on a single substitute. What remains unproven is whether announced capacity can achieve competitive yield, energy efficiency, software compatibility, utilization and customer economics at scale.

## The numbers that anchor the view

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| **92% to 34%**<br>forecast reduction in China's supply-demand gap for logic chips at 7 nm and below between 2025 and 2035, cited from Goldman research in the Daily Brief. [1] | **46% CAGR**<br>forecast growth in China's advanced-process wafer supply from 2025 to 2035, reaching 410,000 wafers per month. [1] |
| **RMB 4.56 trillion**<br>forecast 2030 total addressable market for AI chips in China cited in the same Daily Brief; inference was estimated at about 70%. [1] | **~70%**<br>forecast share of China's 2030 AI-chip market attributed to inference in the cited Daily Brief; this is a forecast from one short-window source. [1] |

## How the signal evolved, 2023-2026

**Multi-year context · Technology fragments**

The CEO archive frames export controls, investment screening and geopolitical risk as persistent strategic variables. Technology access becomes part of resilience planning. [2][3]

**2025-early 2026 · Efficiency and local ecosystems matter more**

External digital and trade research points to continued technology diffusion alongside fragmented standards, infrastructure and supply chains. [4][5]

**August 2026 · The Daily feed shows stack-wide activity**

Daily items cover chips, memory, packaging, cloud, models and financing. The breadth supports a localization hypothesis, but the time window is too short to judge sustainable market share or returns. [6][7][1][8]

## How the trend creates - or destroys - value

**Constraint links the stack**

Restricted access to one layer increases the value of substitutes in adjacent layers. Domestic compute is less useful without memory, packaging, software and power that can deliver a reliable workload. [1][4]

**Efficiency can offset hardware gaps**

Model design, quantization, workload orchestration and software optimization can reduce dependence on the most advanced hardware, although performance and developer costs must be measured in real applications. [7][1]

**Policy accelerates capacity but not necessarily returns**

Public support and strategic demand can fund rapid build-out. Commercial success still depends on yield, utilization, ecosystem compatibility and customer willingness to pay. [6][8]

## Economic logic

The stack should be evaluated through delivered workload economics rather than nominal chip specifications. Relevant measures include yield, energy per useful task, software migration cost, developer productivity, utilization and total cost of ownership. Forecast reductions in supply gaps can coexist with weak economics if capacity is expensive or incompatible with customer workloads. Global companies also face an option problem: supporting multiple stacks raises engineering cost but can preserve market access and resilience. [1][4][9]

## Implications by stakeholder

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| **Chinese technology companies**<br><br>Winning requires coordinated progress across hardware, software, cloud and applications rather than isolated capacity announcements. | **Global vendors**<br><br>Products and roadmaps need explicit scenarios for export controls, local standards and multi-stack support. |
| **Investors**<br><br>Separate strategic importance from commercial return and demand proof on yield, utilization, energy efficiency and customer revenue. | |

## CEO action agenda

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|---|---|
| **NEXT 90 DAYS**<br>**Map stack dependencies**<br>Identify critical dependencies across compute, memory, packaging, software, cloud, power and policy for each product line. | **NEXT 90 DAYS**<br>**Build scenario roadmaps**<br>Define product and sourcing choices under narrow, broader and easing restriction cases. |
| **6-12 MONTHS**<br>**Test multi-stack economics**<br>Measure migration cost, performance, energy and developer productivity on representative workloads. | **6-12 MONTHS**<br>**Verify commercial utilization**<br>Track revenue-bearing deployments and customer retention separately from announced capacity. |

## What could break the thesis

- Restrictions could ease or narrow, reducing the economic value of some localization investments.
- Domestic volume may rise without closing performance, yield or ecosystem gaps.
- Forecast market size may overstate monetizable demand if inference prices fall rapidly.

## Indicators to monitor

- Advanced-process yield and energy efficiency
- Revenue-bearing utilization
- Software migration cost
- Domestic versus imported dependency by layer
- Scope of export and investment controls

## Evidence boundary and confidence

Medium on localization direction; low-medium on the cited 2030-35 market forecasts
This is an emerging signal. The quantitative forecasts come from one Daily Brief citing external research and require continued verification; ten Daily editions cannot establish a structural outcome by themselves.

# Sources and traceability

[1] Weekly Brief · 2026-08-26 · AI□□□Agent□□□□□□□□□□□□□□□□□

[2] Weekly Brief · 2024-06-24 · How to Face Rising Geopolitical Risk

[3] Weekly Brief · 2025-01-14 · Navigating the New Dynamics of Global Trade

[4] Authoritative research · OECD · 2024-11-19 · OECD Digital Economy Outlook 2024 (Volume 2): Strengthening Connectivity, Innovation and Trust

[5] Authoritative research · WTO · 2025-12-15 · Global Value Chain Development Report 2025: Rewiring GVCs in a Changing Global Economy

[6] Weekly Brief · 2026-08-10 · AI□□□□□□□□□□□□□□□□□□□□

[7] Weekly Brief · 2026-08-17 · □□□□AI□□□□□□□□□□□□□□□□□□□□□□□

[8] Weekly Brief · 2026-08-27 · □□□□□□□□□□□□□□AI□□□□□□□□□□□□

[9] Weekly Brief · 2026-07-28 · The AI Bill Lands on the CEO’s Desk

[10] Weekly Brief · 2026-08-13 · AI□□□□□□□□□□□□□□□□□□□□□

[11] Weekly Brief · 2026-08-11 · AI□□□□□□□□□□□□□□□□□□□□

[12] Authoritative research · ITU · 2025-11-27 · Global Connectivity Report 2025

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